



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	INDUSTRIAL CHEMISTRY				
Type of Course	ELECTIVE IN MAJOR				
Semester	VI				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Prior understanding of basic chemistry principles and chemical reactions. 2. Introductory knowledge of industrial processes and terminology.				
Course Summary	The course provides a comprehensive overview of chemical industries, covering industrial processes, waste management, petrochemicals, pharmaceuticals, fertilizers, and Kerala's chemical industries. Students learn about water treatment, safety measures, and the production of synthetic petrol, pharmaceuticals, and fertilizers. By examining various industries, students gain valuable insights into their operations, preparing them for roles in the chemical sector.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Comprehend industrial requirements, including water treatment methods, waste management, and safety protocols.	U	F	Instructor-created exams / Quiz /Assignment
CO2	Investigate natural gas, coal, and crude oil composition, as well as the distillation processes involved	An	C	Class test /Assignment /Quiz
CO3	Analyse drug classifications, terminology, and the preparation of common drugs like paracetamol and aspirin.	An	C	Assignment/ Class test

CO4	Evaluate the production methods of nitrogenous, phosphatic, and potash fertilizers, including NPK fertilizers.	E	P	Assignments /Seminar presentation
CO5	Scrutinize chemical industries in Kerala, focusing on their location, raw materials, and the chemistry involved in product preparation	E	M	Assignments /Seminar presentation
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Introduction (12 hrs)		12	20
	1	Requirements of an industry – location – water – industrial water treatment.	2	
	2	Water softening methods: Clark’s process - lime soda process - Ion exchange process)	2	
	3	Safety measures – pilot plants – ISO certification.	2	
	4	Solid waste management -incineration method, Composting process, disposal.	2	
	5	Liquid waste management- Dewatering, sedimentation, Root-Zone Treatment.	2	
	6	Gaseous waste management-absorption, adsorption, combustion.	2	
II	Petrochemical Industry (12 hrs)		12	20
	7	Introduction. Natural gas – CNG, LNG and LPG. Coal: Classification based on carbon content – carbonisation of coal – composition and uses of various fractions.	2	
	8	Crude Oil: Constitution and distillation – composition and uses of different distillates – ignition point, flash point and octane number – cracking.	2	
	9	Catalysts used in Petroleum Industries: Structure, selectivity and applications. Synthetic Petrol: Manufacture by Bergius and Fischer-Tropsch processes.	3	

	10	Manufacture of petrochemicals: Ethylene glycol, glycerine, acetone, phenol, vinyl acetate, toluene, linear alkyl benzenes and their sulphonates.	3	
	11	Usage and depletion of petroleum products – need for alternative fuel – hydrogen as the future fuel.	2	
III	Pharmaceutical Industry (12 hrs)		12	20
	12	Drugs: Definition – History of drugs – Terminology: Prodrug, pharmacy, pharmacology, pharmacodynamics and pharmacokinetics (elementary idea only).	2	
	13	Antipyretics, analgesics and antacids (definition and examples, structures not expected)	2	
	14	Antihistamines, antibiotics, antiseptics and disinfectants, (definition and examples, structures not expected)	2	
	15	Anti-inflammatory agents, Sedatives, Tranquilizers, Hypnotics and Antidepressant drugs (definition and examples, structures not expected) – Preparation of paracetamol and aspirin.	3	
	16	Drug toxicity – Thalidomide tragedy (a brief study) – Effective use of drugs – Over dosage – Prescription and non-prescription drugs – Definition, examples, uses and side effects	2	
	17	Drug abuse- Medical applications of metal and metal oxide nanomaterials.	1	
IV	Fertilizer Industry (12 hrs)		12	20
	18	Introduction- Nitrogenous, phosphatic and potash fertilizers, NPK fertilizers, NPK value, Manufacturing methods of ammonium nitrate- Prilling method, Stengel method.	3	
	19	Urea–Manufacture from ammonia and carbon dioxide.	1	
	20	Monoammonium Phosphate (MAP) and Diammonium Phosphate (DAP)- Manufacture from ammonia and phosphoric acid.	2	
	21	Potassium Chloride (muriate of potash)- main steps involved in the manufacture. Mining of the K mineral, Separation of the main ingredient and purifying.	3	
	22	Potassium Sulphate (sulfate of potash) - Manufacture from langbeinite (K ₂ SO ₄ . MgSO ₄) and KCl. NPK (17-17-17)- Granulation method of manufacture.	3	
V	Chemical industries in Kerala: (12 hrs)		12	20
	Open ended	Location, raw materials, chemistry involved in the preparation and uses of the following, caustic soda and chlorine – Travancore	12	

	.	Cochin Chemicals Ltd., TiO ₂ pigment from ilmenite – Travancore Titanium Products Ltd.		
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References:

1. Industrial Chemistry (including chemical - engineering) - B.K. Sharma - Goel publishing house, Meerut.
2. Pollution control in process industries - S.P. Mahajan - Tata McGraw - Hill Publishing Company Ltd., New Delhi.
3. Water pollution and management - C.K. Varashney - Wiley Eastern Ltd., Chennai - 20
4. E. Stocchi, Industrial Chemistry, Vol. I, Ellis Horwood Ltd. UK, 1990.
5. P. C. Jain, M. Jain, Engineering Chemistry, Dhanpat Rai & Sons, Delhi, 2015.
6. B. K. Sharma, H. Gaur, Industrial Chemistry, Goel Publishing House, Meerut, 1996.
7. G. L. Patrick, Introduction to Medicinal Chemistry, 6 th Edn., Oxford University Press, UK, 2017.
8. Hakishan Singh, V. K. Kapoor, Medicinal and Pharmaceutical Chemistry, Vallabh Prakashan, Pitampura, New Delhi, 2005.
9. Thomas L. Lemke, David A. William, Foye's Principles of Medicinal Chemistry, 6 th Edn., Wolters Kluwer Health, 2006.
10. Jayashree Ghosh, A Text Book of Pharmaceutical Chemistry, S. Chand and Co. Ltd, 1999.
11. O. Le. Roy, Natural and synthetic organic medicinal compounds, Ealemi, 1976.
12. . Hand book on fertilizer technology fertilizer association of india near JNU New Delhi 1992
13. Sauchelli V. The chemistry and technology of fertilizers ACS Monograph no.148, Reinhold publishing Cor. NewYork 1980

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	1						3		2		2		1
CO 2	2						3		2		2		1
CO 3					1	1	3		2		2		1
CO 4						1	3		2		3		12
CO 5						1	3		1		2		1

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓